Mahdi Alam

Toronto, ON | 647-916-4783 | mahdi.alam@torontomu.ca
mahdialam.com | github.com/mahdi-alam360 | linkedin.com/in/mahdi-alam-264127150

PROFILE

Computer Science graduate and MSc researcher with six years of Python experience, computer-vision experimentation and web-project experience. Built an AI-assisted C#/ASP.NET Core imagery-review prototype with SQLite, parameterized SQL, request validation and tested persistence. Interested in image analytics, reliable web services and careful software testing.

EDUCATION

Toronto Metropolitan University | Toronto, ON

MSc, Computer Science | Thesis research in artificial intelligence
September 2023 - Expected December 2026 (possible extension to January 2027)

BSc (Honours), Computer Science | September 2019 - June 2023

TECHNICAL SKILLS

Programming: Python, C++, C; C# and JavaScript in personal/academic projects

Web and databases: ASP.NET Core and SQLite (new learning project); React, TypeScript, Node.js; PHP, MySQL and SQL project experience

ML and tools: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, SciPy, OpenCV, Git, Docker, Linux

RESEARCH EXPERIENCE

MSc Thesis Research - Self-Supervised Medical Image Registration
Toronto Metropolitan University | Toronto, ON

- Designed a two-phase, coarse-to-fine training strategy combining multi-resolution curriculum learning and adaptive optimization for self-supervised deep learning.
- Improved Dice Similarity Coefficient by up to 1.7 points across public medical-image datasets (LPBA40, IXI, OASIS), using quantitative benchmark evaluation.
- Evaluated deformation quality through Jacobian determinant analysis and conducted reproducible Python experiments with ablation studies.

SELECTED PROJECTS

RoadFrame - Imagery Review Lab | C#, ASP.NET Core, SQLite | October 2026

AI-assisted personal learning project | mahdialam.com/roadframe/

- Built a local REST API for synthetic road-scene reviews with SQLite persistence, parameterized queries, foreign keys, upserts and JOIN/GROUP BY/AVG summary reports.
- Verified 18 HTTP/SQL integration assertions, including invalid-input rejection, updates, injection-shaped notes and persistence after restart. Published source and a browser-only interactive demo; no production deployment or automated vision-model claims.

Sofa Store Website | React, AngularJS, JavaScript, PHP, MySQL | Jan-May 2022

- Developed a storefront with user accounts, MVC structure and MySQL-backed user/product data; integrated Google Maps API. Academic/personal project, not industry employment.

Document Search Lab | React, TypeScript, Node.js | October 2026

- Built an AI-assisted BM25 search interface/API; verified 15 automated ranking, validation and HTTP tests, TypeScript checks and build.

Search Engine - NLP and Retrieval | Python | October-November 2022

- Implemented TF-IDF/BM25 retrieval with Naive Bayes and linear SVM baselines; compared retrieval effectiveness.

AVAILABILITY

Canadian citizen; authorized to work in Canada without sponsorship. Willing to relocate within Canada. Available one month after an offer. MSc expected December 2026, possibly January 2027. Research manuscript in preparation, not published.